Inference and effect size

Reminders

- Exam 1 next Friday (10/10)
    - Review day next Wednesday (10/8)
- Exam 1 material: everything through the end of this week (10/3)
    - I will not make you write any R code
    - You will need to read and interpret R output
    - Bring a scientific calculator! Phones are not allowed.
- Studying tips:
    - Practice questions on the course website
    - Look back at class activities
    - Go over homework questions

Last time: Activity

```
              Estimate Std. Error z value Pr(>|z|)    
(Intercept) -2.507329   0.311939  -8.038 9.14e-16 ***
loss        -0.846499   0.238335  -3.552 0.000383 ***
smoke100    -0.137151   0.035816  -3.829 0.000128 ***
hlthplan     0.613613   0.051233  11.977  < 2e-16 ***
age         -0.012148   0.001043 -11.652  < 2e-16 ***
height       0.056845   0.004520  12.577  < 2e-16 ***
```

The standard deviation of the `loss` variable is 0.074.

> How do the odds of exercise change with a one standard deviation increase in loss, holding the other variables constant?

Activity

```
             Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.507329   0.311939  -8.038 9.14e-16 ***
loss        -0.846499   0.238335  -3.552 0.000383 ***
...           ...         ...        ...     ...
```

The standard deviation of the `loss` variable is 0.074.

> Want to test whether a one-standard deviation increase in `loss` is associated with a decrease in the odds of at least 5% (holding other variables constant).

Statistical "significance" vs. practical significance

- A small p-value in the test of $H_0 : \beta = 0$ is evidence against H_0
- However, we could have $\beta \neq 0$ but still *very close* to 0
- Small p-values therefore do not mean an effect has *practical* significance
- It is common for domain experts to specify what effect size would be of practical significance, and then we can test whether the effect is at least that big

Activity

Work on the class activity (handout), then we will discuss as a group.

Activity

```
Coefficients:
               Estimate Std. Error z value Pr(>|z|)
(Intercept)   -3.989979   1.139951  -3.500 0.000465 ***
gre            0.002264   0.001094   2.070 0.038465 *
gpa            0.804038   0.331819   2.423 0.015388 *
factor(rank)2 -0.675443   0.316490  -2.134 0.032829 *
factor(rank)3 -1.340204   0.345306  -3.881 0.000104 ***
factor(rank)4 -1.551464   0.417832  -3.713 0.000205 ***
```

> Given your friend's GPA (3.95), school (rank 3), and current GRE score (590), what is your estimated probability that they will be accepted to graduate school?

Activity

```
Coefficients:
                 Estimate Std. Error z value Pr(>|z|)
(Intercept)    -3.989979   1.139951  -3.500 0.000465 ***
gre             0.002264   0.001094   2.070 0.038465 *
gpa             0.804038   0.331819   2.423 0.015388 *
factor(rank)2 -0.675443   0.316490  -2.134 0.032829 *
factor(rank)3 -1.340204   0.345306  -3.881 0.000104 ***
factor(rank)4 -1.551464   0.417832  -3.713 0.000205 ***
```

> The standard deviation of GRE scores is 115.5 points. If your friend re-took the GRE and increased their score by one standard deviation, what would be the estimated change in their *odds* of being accepted?

Activity

```
Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)    -3.989979   1.139951  -3.500 0.000465 ***
gre             0.002264   0.001094   2.070 0.038465 *
gpa             0.804038   0.331819   2.423 0.015388 *
factor(rank)2 -0.675443   0.316490  -2.134 0.032829 *
factor(rank)3 -1.340204   0.345306  -3.881 0.000104 ***
factor(rank)4 -1.551464   0.417832  -3.713 0.000205 ***
```

> If your friend re-took the GRE and increased their score by one standard deviation, what would be their new estimated *probability* of getting accepted?

Activity

```
Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)    -3.989979   1.139951  -3.500 0.000465 ***
gre             0.002264   0.001094   2.070 0.038465 *
gpa             0.804038   0.331819   2.423 0.015388 *
factor(rank)2 -0.675443   0.316490  -2.134 0.032829 *
factor(rank)3 -1.340204   0.345306  -3.881 0.000104 ***
factor(rank)4 -1.551464   0.417832  -3.713 0.000205 ***
```

> Calculate a 95% confidence interval for the change in *odds* of acceptance associated with a *one-unit* increase in GRE score, holding GPA and school rank fixed

Activity

```
Coefficients:
                Estimate Std. Error z value Pr(>|z|)
(Intercept)    -3.989979   1.139951  -3.500 0.000465 ***
gre             0.002264   0.001094   2.070 0.038465 *
gpa             0.804038   0.331819   2.423 0.015388 *
factor(rank)2 -0.675443   0.316490  -2.134 0.032829 *
factor(rank)3 -1.340204   0.345306  -3.881 0.000104 ***
factor(rank)4 -1.551464   0.417832  -3.713 0.000205 ***
```

> Now calculate a 95% confidence interval for the change in *odds* of acceptance associated with a one *standard deviation* increase in GRE score, holding GPA and school rank fixed

Activity

95% confidence interval for change in *odds* of acceptance associated with a one *standard deviation* increase in GRE score, holding GPA and school rank fixed:

$$(1.014, 1.66)$$

Your friend is reluctant to retake the exam unless they believe that their odds of acceptance will increase by at least 25%.

> Would you recommend that your friend re-takes the exam?